

Chapter 5 Phase diagram of Ceramics

—— Gibbs Phase Rule

- Phase diagrams and phase equilibria are subject to the laws of thermodynamics.
- Gibbs phase rule is a criterion that determines how many phases can coexist within a system at equilibrium.

$$P + F = C + N$$

P: # of phases present

F: degrees of freedom (temperature, pressure, composition)

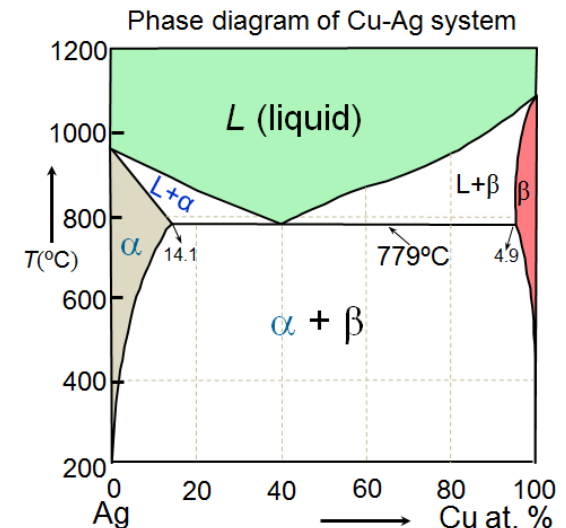
C: components or compounds

N: noncompositional variables

For the Cu-Ag system @ 1 atm for a single phase P:

N=1 (temperature), C = 2 (Cu-Ag), P= 1 (α , β , L)

$F = 2 + 1 - 1 = 2$



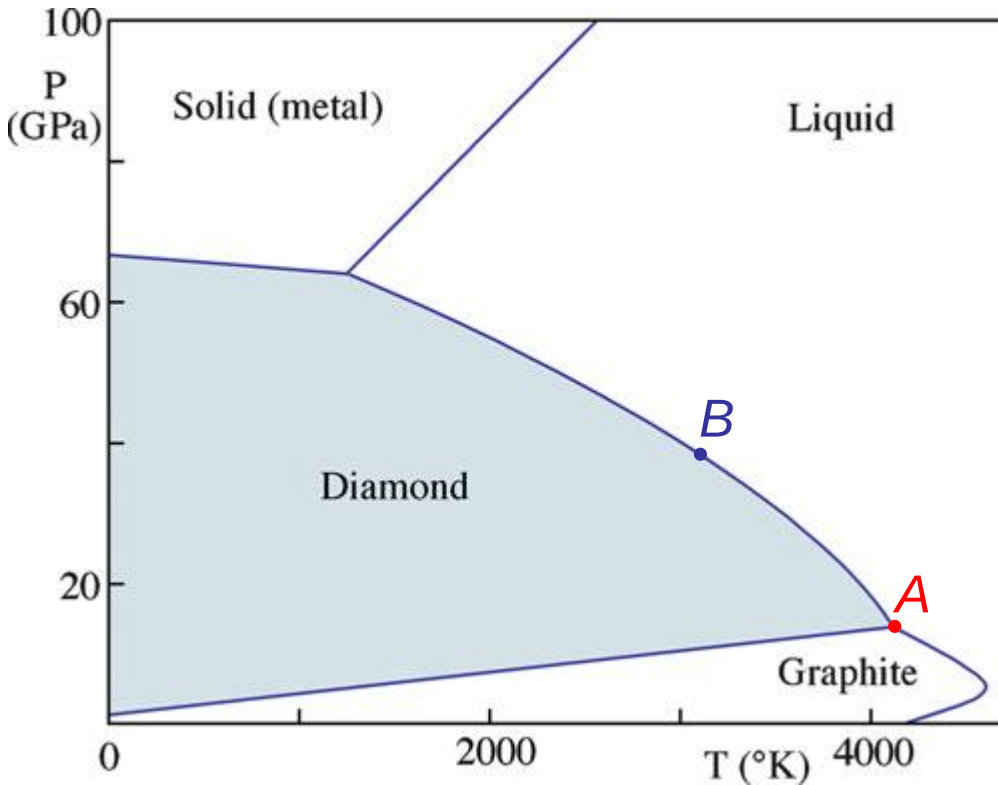
This means that to characterize the alloy within a single phase field, 2 parameters must be given: temperature and composition.

If 2 phases coexist, for example, α +L, β +L, α + β , then according to GPR, we have 1 degree of freedom: $F = 2 + 1 - 2 = 1$. So, if we have Temp or composition, then we can completely define the system.

If 3 phases exist (for a binary system), there are 0 degrees of freedom. This means the composition and Temp are fixed. This condition is met for a eutectic system by the eutectic isotherm.

Single System (C=1)

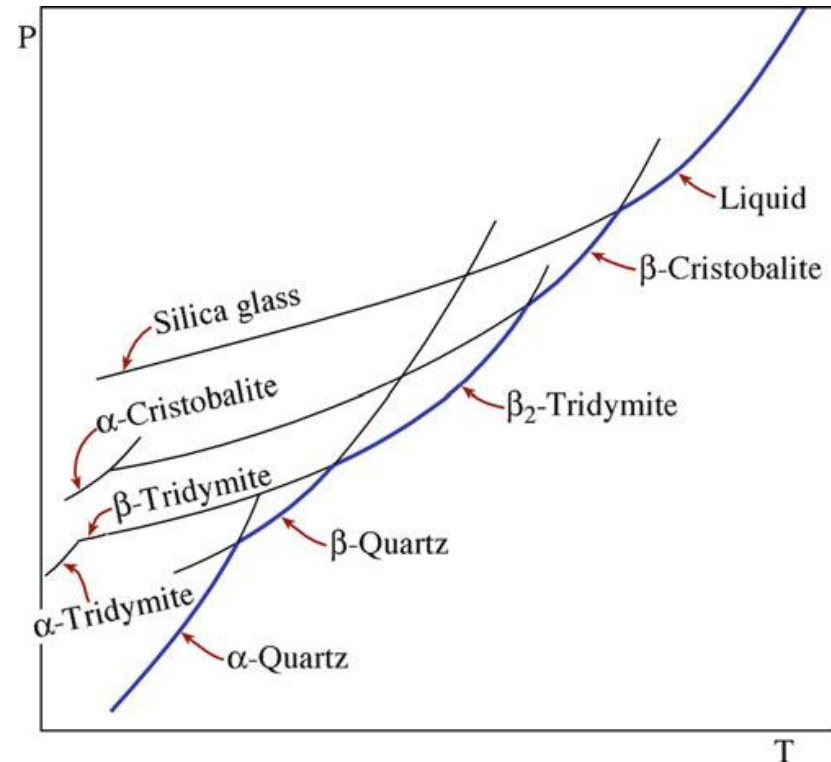
C Phase Diagram



A: C=1, P=3, F=0

B: C=1, P=2, F=1

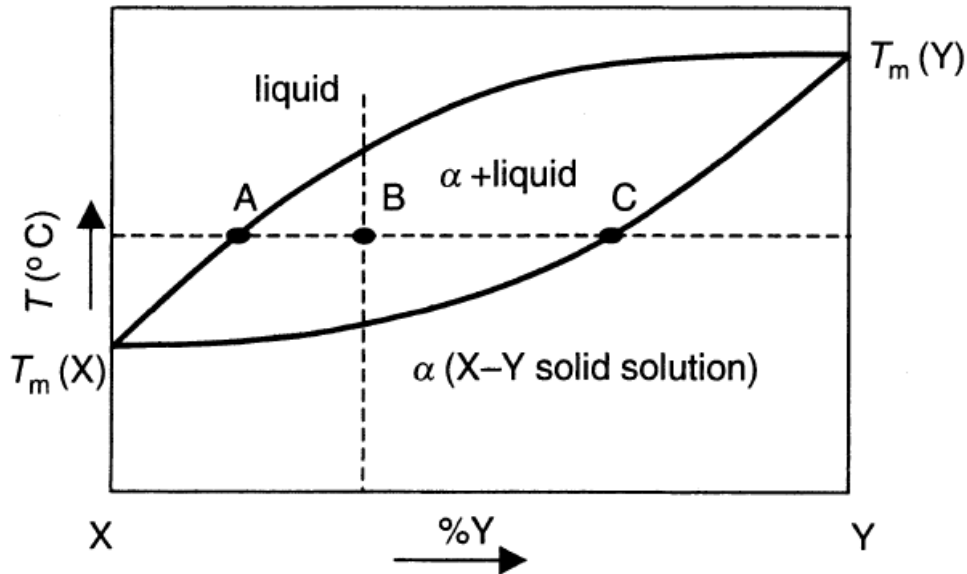
SiO₂ Phase Diagram



Polymorphic Transformations

Binary Systems (C=2)

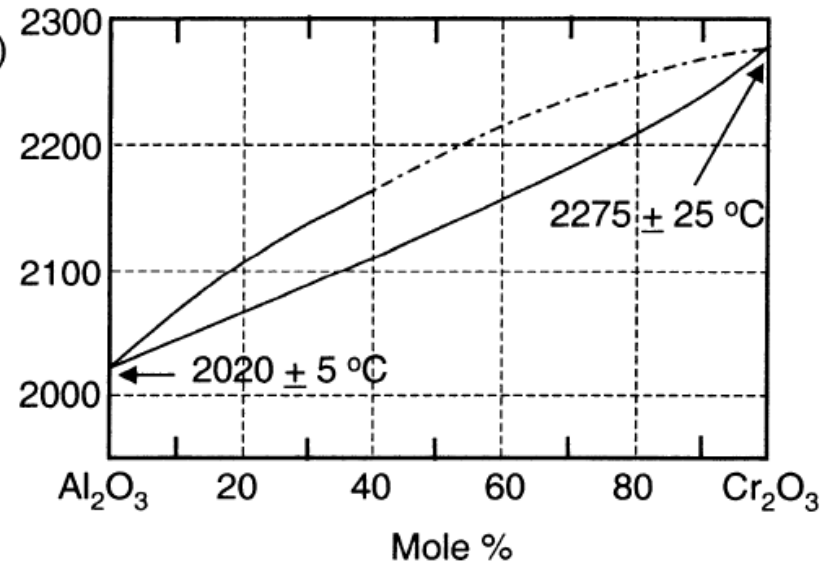
Isomorphous System



$$\% \text{ Liquid} = (BC/AC) \times 100$$

$$\% \text{ Solid} = (AB/AC) \times 100$$

Al_2O_3 - Cr_2O_3 diagram

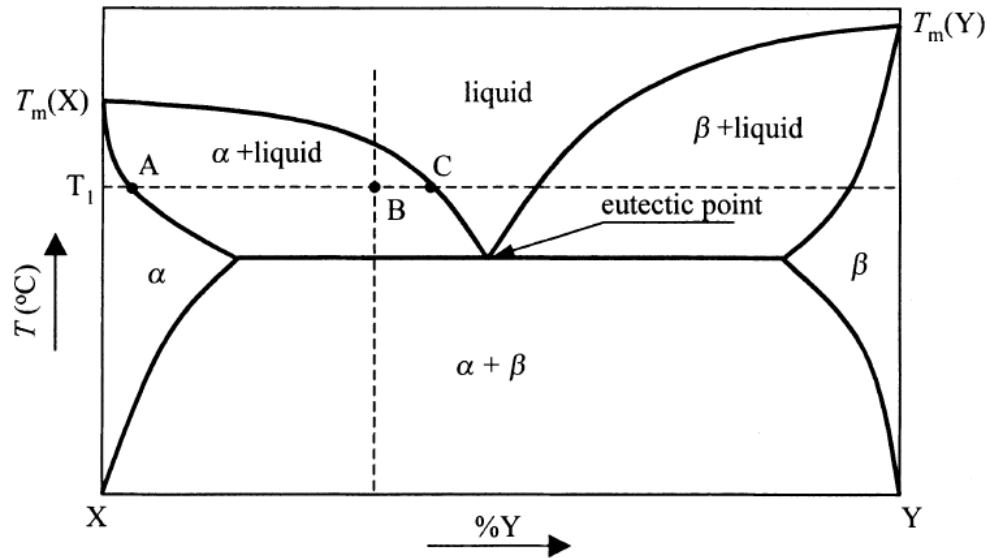


Complete Solubility

- 1) Same type of Lattice Structure
- 2) Same valence
- 3) Small size difference between ions ($< 15\%$)

Binary Systems (C=2)

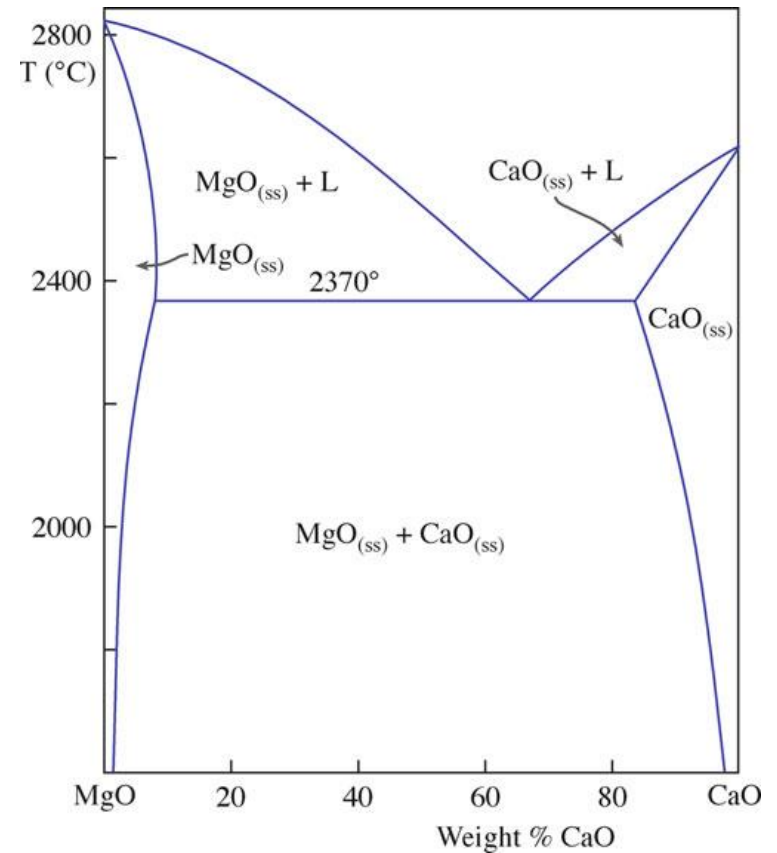
Eutectic phase diagram



$$\% \text{ Liquid} = (AB/AC) \times 100$$

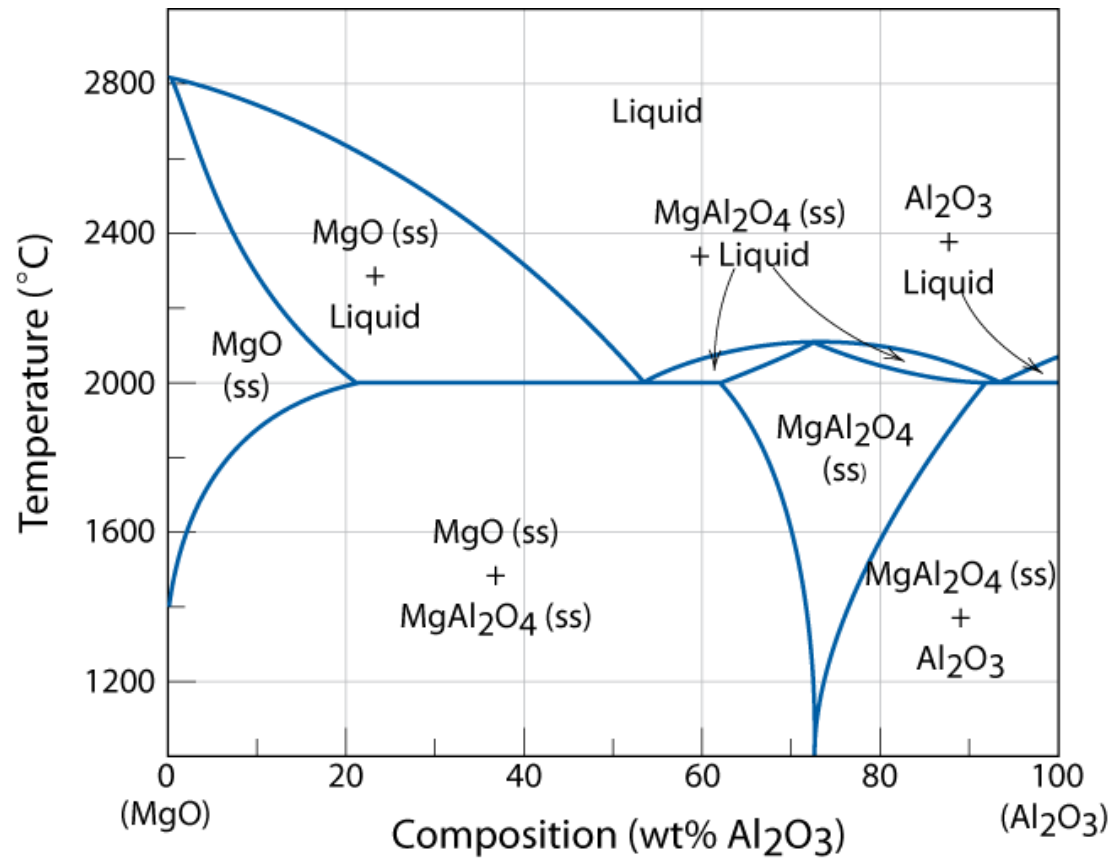
$$\% \alpha = (BC/AC) \times 100$$

MgO-CaO diagram



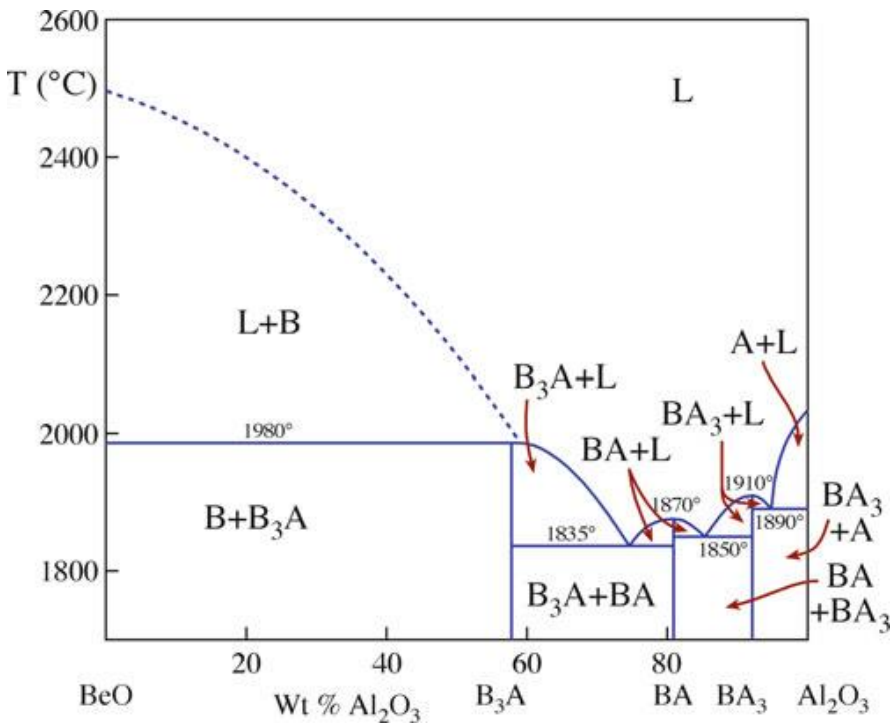
Binary Systems (C=2)

MgO-Al₂O₃ diagram

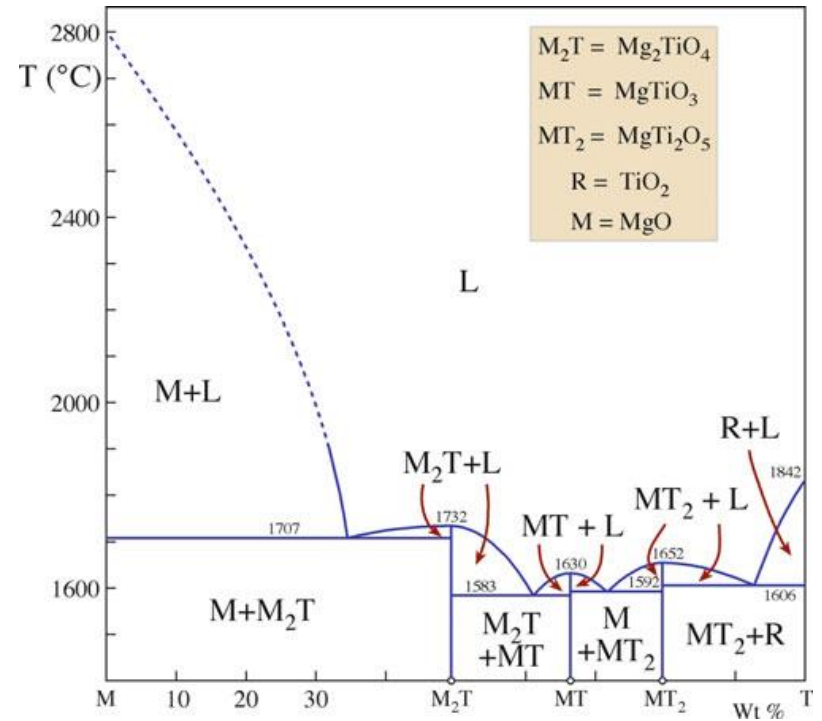


Binary Systems (C=2)

BeO-Al₂O₃ diagram

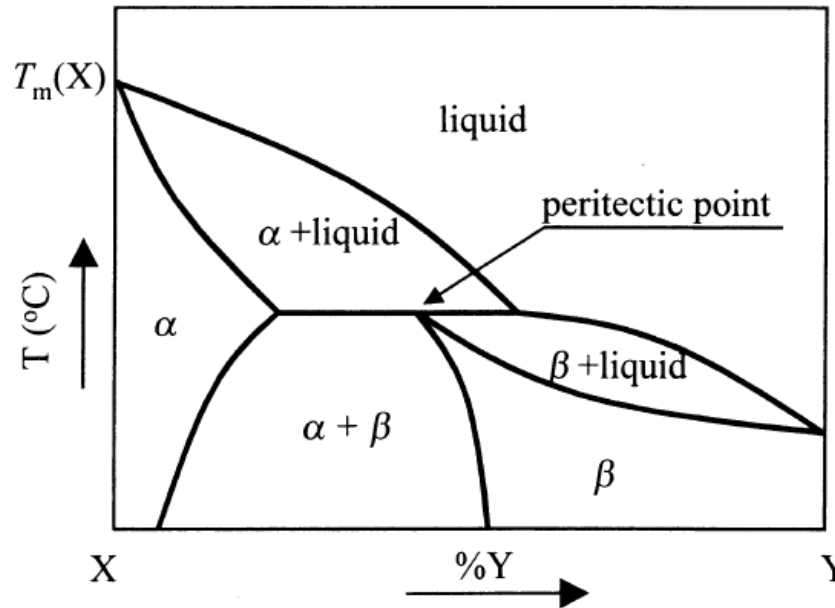


MgO-TiO₂ diagram



Binary Systems (C=2)

Eutectic phase diagram

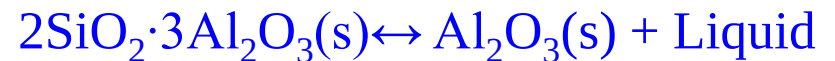
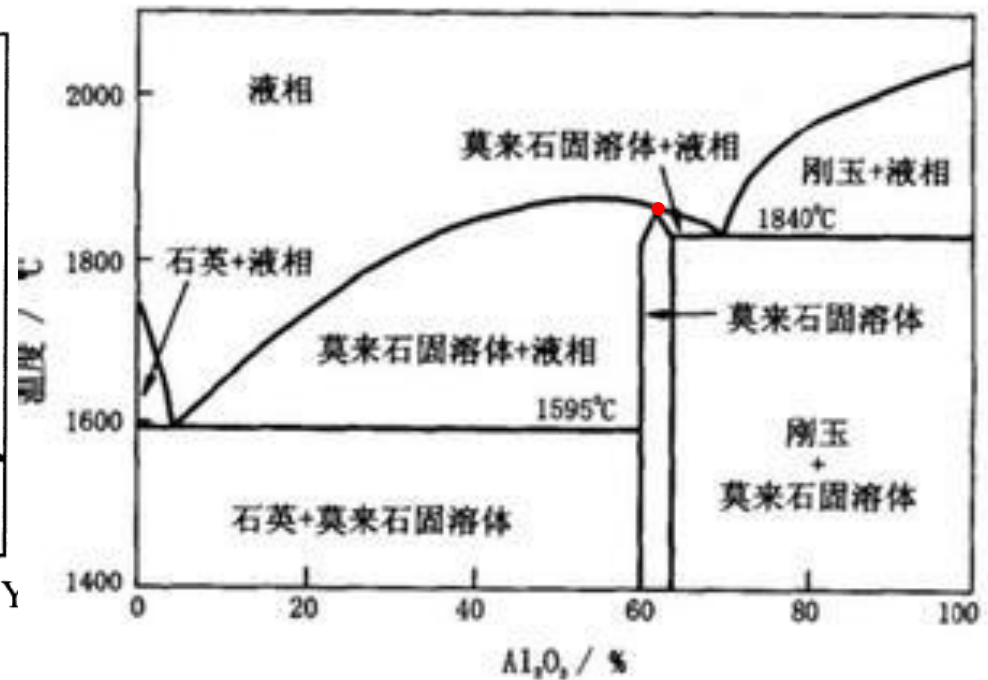


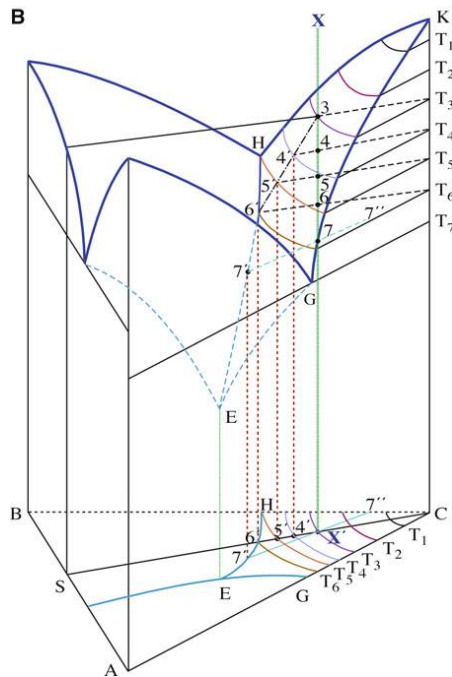
$$\% \text{ Liquid} = (\text{AB}/\text{AC}) \times 100$$

$$\% \alpha = (\text{BC}/\text{AC}) \times 100$$



Silica-Alumina diagram



MgO-Al₂O₃-SiO₂ diagram

How to Read Phase diagram of Ternary Systems ?

Since $oa = bC = W_A$

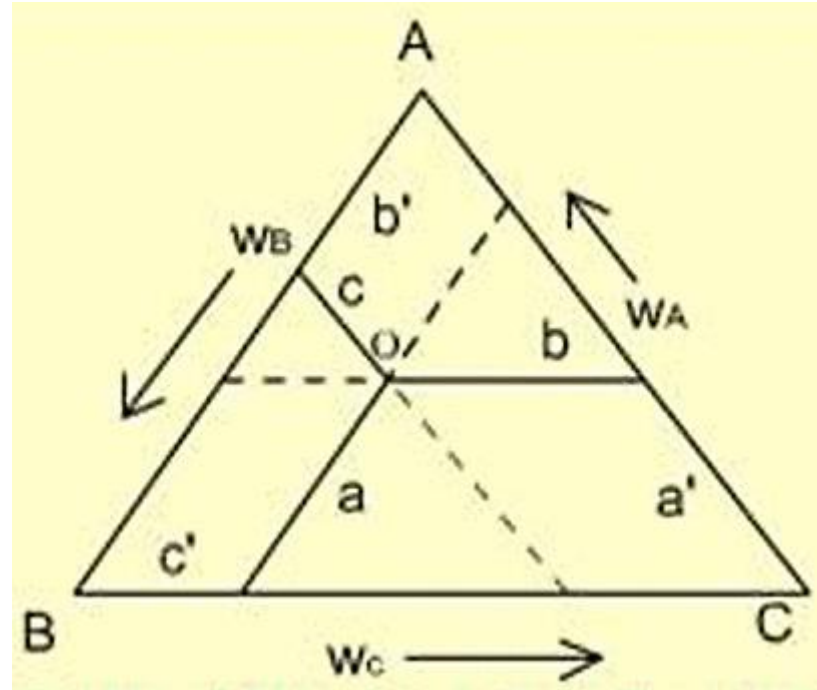
$ob = Ac = W_B$

$oc = Ba = W_C$

$oa \rightarrow A \text{ wt}\%$,

$ob \rightarrow B \text{ wt}\%$,

$oc \rightarrow C \text{ wt}\%$.



How to Read Phase diagram of Ternary Systems ?

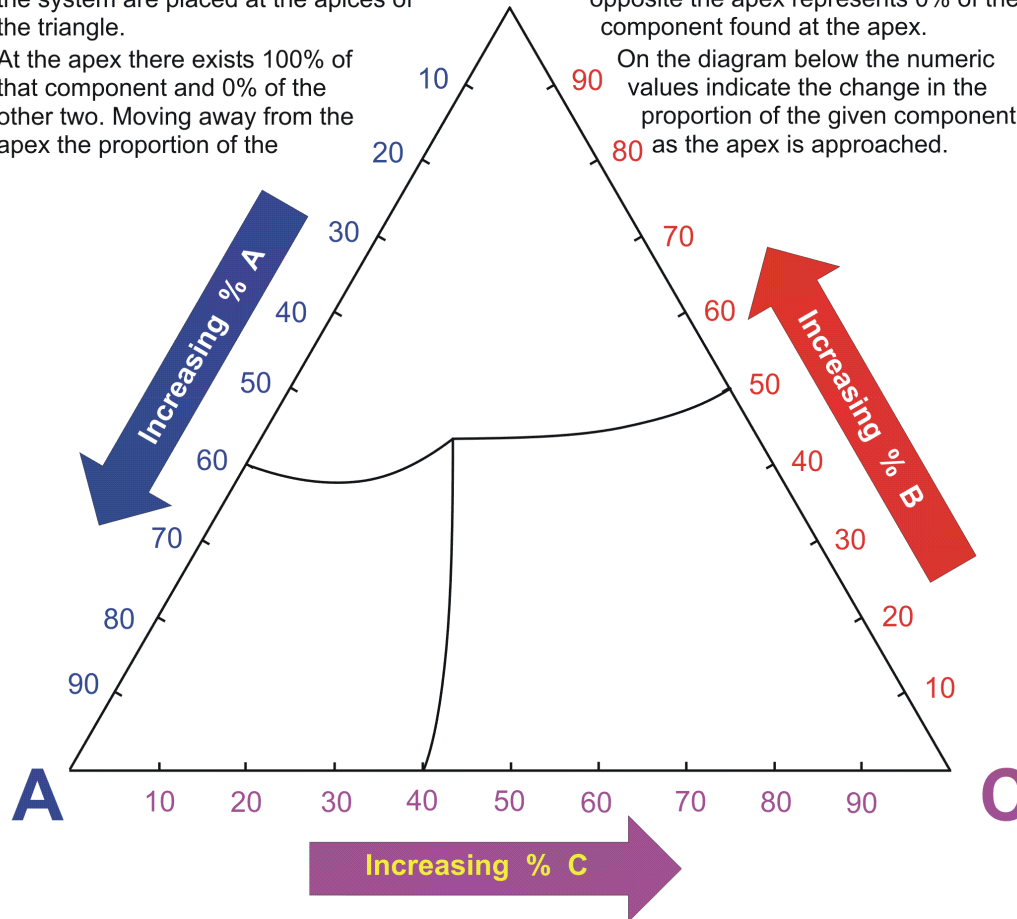
Ternary diagrams are most commonly plotted as equilateral triangles. The three components (**A**, **B** and **C**) which define the system are placed at the apices of the triangle.

At the apex there exists 100% of that component and 0% of the other two. Moving away from the apex the proportion of the

B

component at the apex decreases as the proportion of the remaining two increase, such that the side of the triangle directly opposite the apex represents 0% of the component found at the apex.

On the diagram below the numeric values indicate the change in the proportion of the given component as the apex is approached.

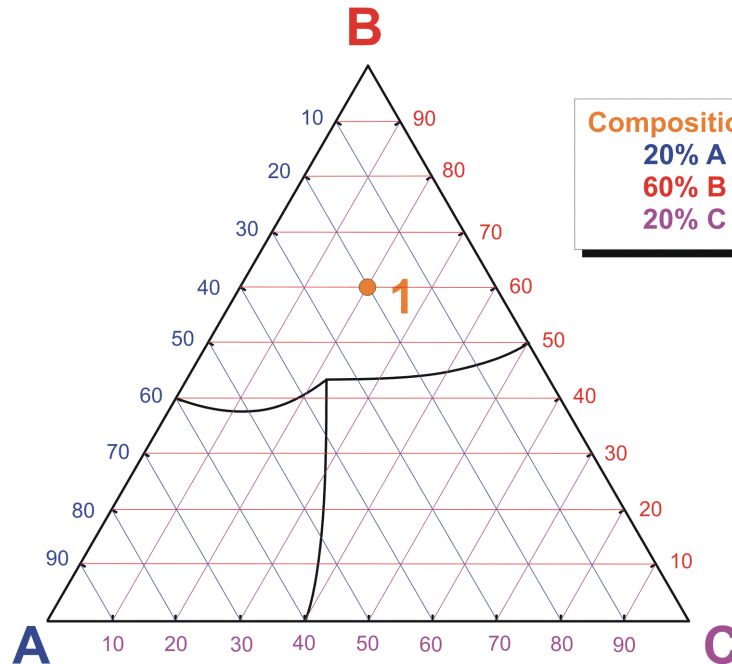


How to Read Phase diagram of Ternary Systems ?

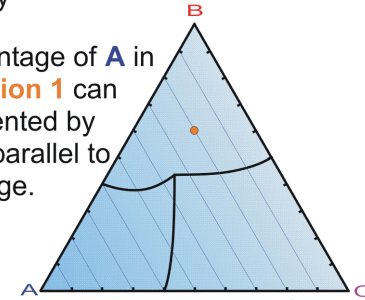
Compositions of points which lie inside the ternary diagram can be determined by using two methods: 1) Triangular Grid or 2) The Two Line Method.

Method 1 - Triangular Grid

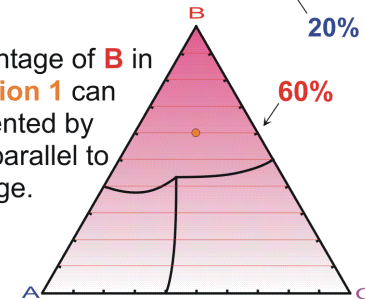
In this method a 10% triangular grid is constructed. The proportion of any point (e.g. **Composition 1**) within the triangle can be represented by grid lines drawn through the point of interest, parallel to each side of the triangle.



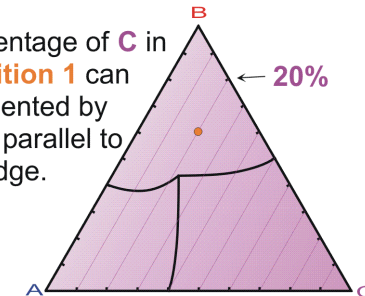
The percentage of **A** in **Composition 1** can be represented by **grid lines** parallel to the **BC** edge.



The percentage of **B** in **Composition 1** can be represented by **grid lines** parallel to the **AC** edge.



The percentage of **C** in **Composition 1** can be represented by **grid lines** parallel to the **AB** edge.



How to Read Phase Diagram of Ternary Systems ?

Method 2 - Two Line Method

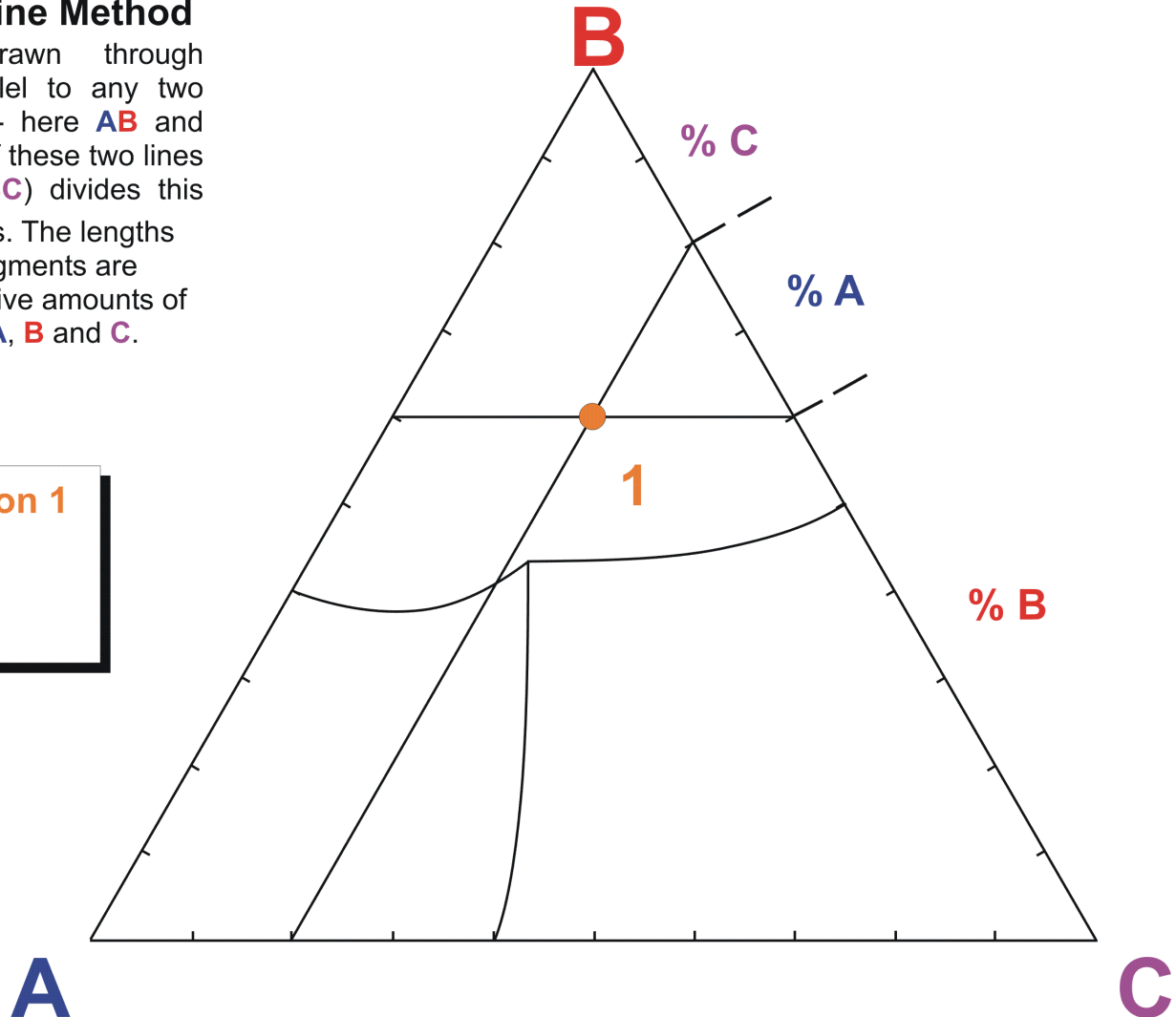
Two lines are drawn through **Composition 1** parallel to any two sides of the triangle - here **AB** and **AC**. The intersection of these two lines with the third side (**BC**) divides this side into three line segments. The lengths of the individual line segments are proportional to the relative amounts of the three components **A**, **B** and **C**.

Composition 1

20% A

60% B

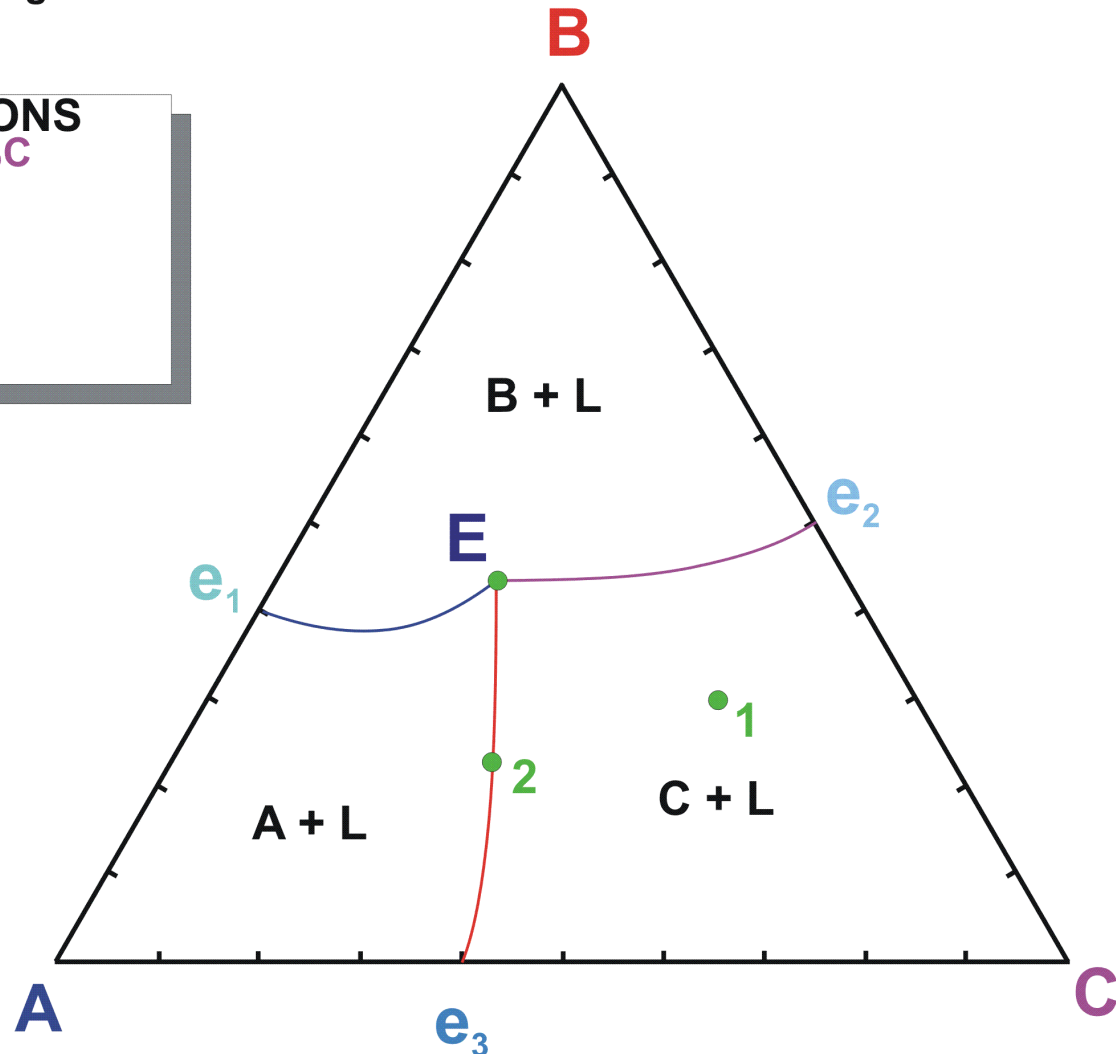
20% C



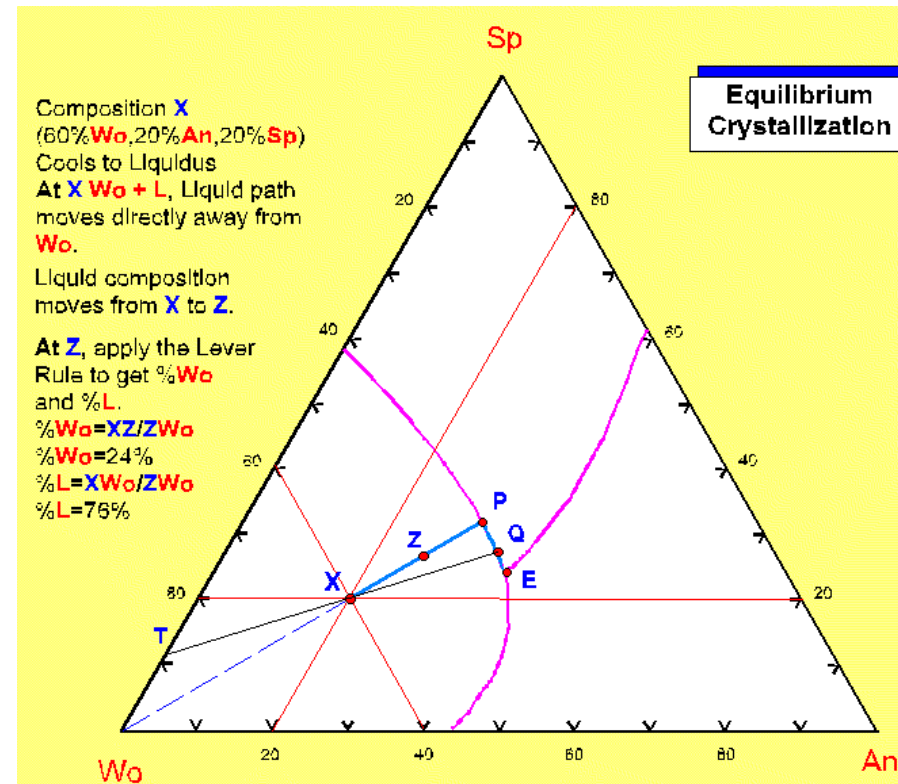
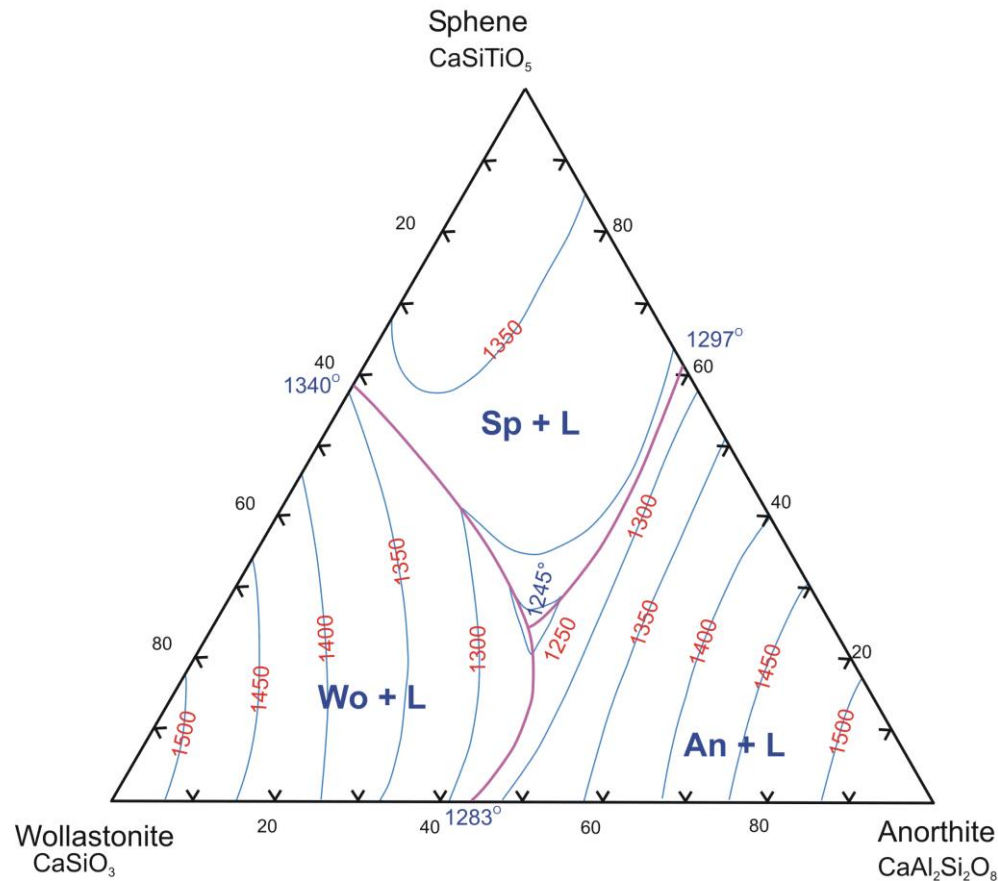
Composition Determination of Ternary Phase Diagram

Determine the compositions of the points in the following table.

COMPOSITIONS		
% A	% B	% C
E		
1		
2		
e ₃		

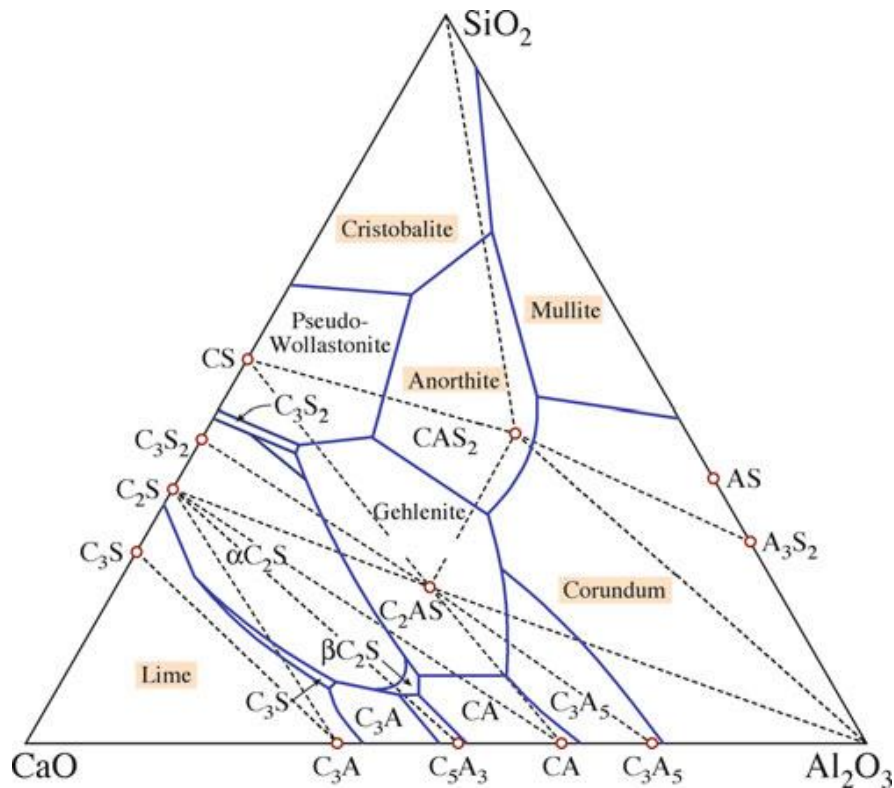


Ternary Systems (C=3)

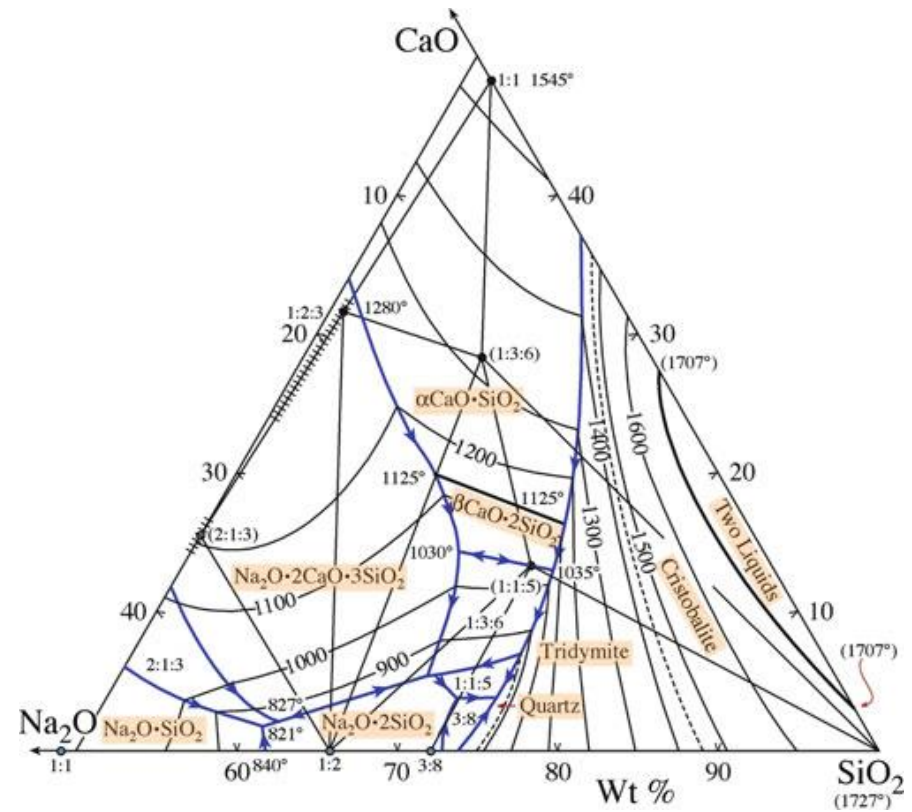


Ternary Systems (C=3)

CaO-SiO₂-Al₂O₃ diagram

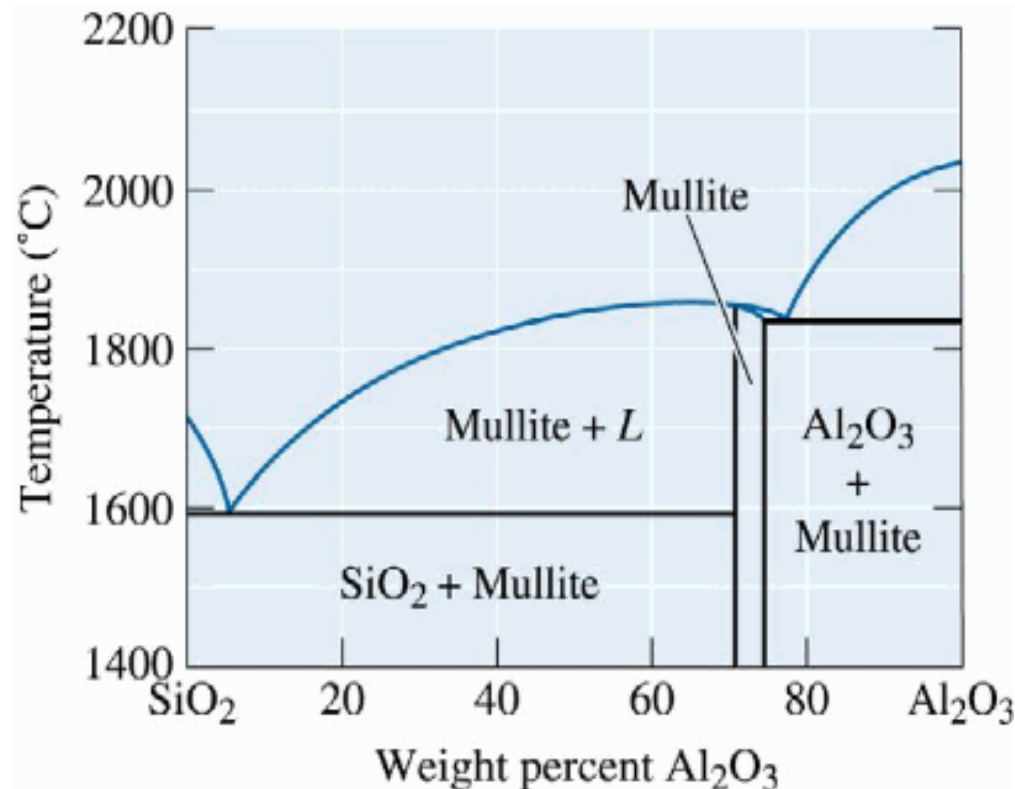


Na₂O-CaO-SiO₂ diagram



Application: Refractories

- Need a material to use in high temperature furnaces.
- Consider Silica (SiO_2) - Alumina (Al_2O_3) system.
- Phase diagram shows: **mullite**, alumina and **crystobalite** (made up of SiO_2) are candidate refractories.



Ceramic Phases and Cements

